

LESSON 5-5 • SESSION 2 • PRACTICE • FORM A

Missing Fractional Edges

Learning target – I can find a missing edge of a prism and find volumes with fractional edge lengths. • 6.G.A.2

REMEMBER

- $V = l \times w \times h$ works with fractions: $12 \times 10 \times 3/4 = 120 \times 3/4 = 90$.
- $1\frac{1}{2}$ of a number is the number **plus half** of it.
- **Missing edge:** multiply the two known edges, then **divide** the volume by that product.

WORKED EXAMPLE

A box holds 72 cm^3 . It is 6 cm long and 4 cm wide.**Step 1** Known edges: $6 \times 4 = 24\text{ cm}^2$.**Step 2** Divide: $72 \div 24 = 3$.**Step 3** Check: $6 \times 4 \times 3 = 72$.**Answer height = 3 cm****WORDS** edge (arista) • fraction (fracción) • volume (volumen)

PRACTICE

show your work • label units

1. Find each volume or missing edge.

a. $1/2\text{ ft} \times 3/4\text{ ft} \times 2\text{ ft}$ ____ ft^3

b. $14\text{ in} \times 11\text{ in} \times 1/2\text{ in}$ ____ in^3

c. $V = 336\text{ in}^3$, $w = 8\text{ in}$, $h = 6\text{ in}$; $l =$ ____ in

d. $V = 96\text{ in}^3$, $B = 12\text{ in}^2$; $h =$ ____ in

- 2.
- Shipping Box Limit.**
- A shipping box holds
- 990 in^3
- . It is 9 in wide and 10 in tall. Boxes may be at most 13 in long.

Hint Multiply the two known edges, then divide.

a. Multiply the two known edges. ____ in^2

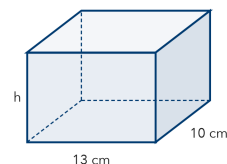
b. Find the length. ____ in

c. Does the box meet the rule? ____

Show your work.

- 3.
- Planter Depth.**
- A planter holds
- $1,170\text{ cm}^3$
- of soil. Its base is 13 cm by 10 cm. How tall is it?

Show your work.

**Answer** _____ cm

- 4.
- Thin Box.**
- A flat box is 20 in long, 16 in wide, and
- $3/4$
- in tall. Find its volume.

Hint Find the base, then take $3/4$ of it.

Show your work.

Answer _____ in^3

- 5.
- Seed Flat.**
- A seed flat is 20 in long, 12 in wide, and
- $1\frac{1}{2}$
- in deep. Find its volume.

Hint $1\frac{1}{2}$ means the base plus half the base.

Show your work.

Answer _____ in^3

6. **Concrete Slab.** A concrete slab is 22 ft long, 9 ft wide, and $\frac{1}{2}$ ft thick. Concrete costs \$5 per ft^3 . Find the volume and the cost.

Hint The thickness is the height.

Show your work.

Volume, cost _____

7. **Aquarium Width.** An aquarium holds $5,400 \text{ in}^3$ of water. It is 30 in long and 18 in tall. The width label wore off. Find the width.

Show your work.

Answer _____ in

8. **Planter Box.** A planter is $\frac{3}{4}$ m long, $\frac{1}{2}$ m wide, and 2 m tall. Soil costs \$18 per m^3 .

- a. Find the volume. _____ m^3
 b. Find the cost to fill it. _____
 c. A second planter is twice as tall. Find its volume. _____ m^3

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Hana’s Work.** A box holds $1,980 \text{ cm}^3$. It is 11 cm wide and 9 cm tall. Hana wrote: “ $1,980 \div 11 = 180$, so the length is 180 cm.” Explain the mistake and find the length.

TEST PRACTICE

choose the best answer

10. A prism has a volume of 120 cm^3 , a length of 5 cm, and a width of 4 cm. Its height is –
A 6 cm **B** 20 cm **C** 24 cm **D** 111 cm
11. A block is $\frac{1}{2}$ m by $\frac{1}{2}$ m by 4 m. Its volume is –
A 1 m^3 **B** 2 m^3 **C** 4 m^3 **D** 5 m^3

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

10 I	9 R	1 1/2 N	495 T	3/4 O	180 Q	240 A	360 C	11 F	110 Z
#2b		#3	#4	#5	#6	#7	#8a	#8c	

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #_____